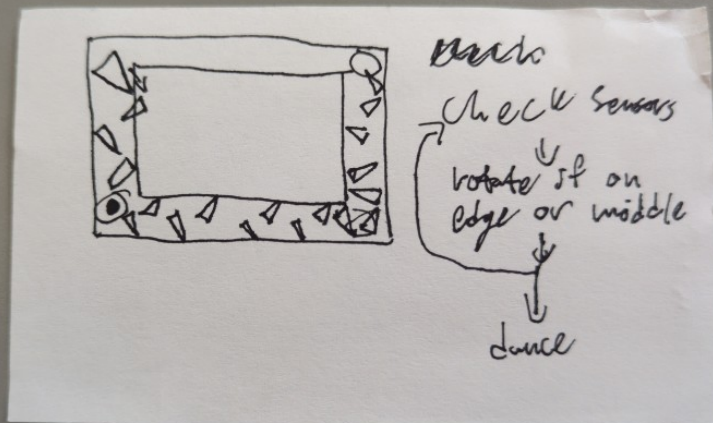


Design Concept Worksheet**Team Name: Evil Robot Wranglers****Design Concept Name: Motion Subsystem Design Concept**

Sketch: In the space below, draw a neat sketch that communicates the navigation and control strategy for your robot. You may want to use multiple sketches to fully communicate your idea.



Narrative Description: Write two to three paragraphs description focused on the key features and principles of operation of the design and how those features work together to support the performance criteria you identified in the problem specification.

This is the main design ideas for the motion subsystem

Step 1, find out where we are

For this step, our main idea is to rotate (and potentially move) until we find the white edge.

Step 2, find the first color

At which point we will move along it. Adjusting course occasionally to stay on the path. Once we find a color, if it is the correct first color, then we continue and do our dance, and go to the next color. If it is the wrong one, we turn around and find the right one.

Step 3, get the rest of the colors

We will continue around the exterior of the track, finding the colors in the right order. Once we have gone to each and done the dance, we will procede to do the victory dance.

Event loop

Each event loop we will do three things.

1. We will check sensors to find out what has changed since the last event cycle.
2. We will do calculations to figure out what we need to do next.
3. We will command the motors to move according to the new trajectory.

Algorithm: In the space below, write pseudocode (plain English summary of the MicroPython code you will use) and/or draw a flow chart that communicates the basic structure of the code you will write to implement the strategy you described above. Include all the inputs and outputs in your system design. This pseudocode or flow chart should explain how the system will respond to various sensor readings.